

ABOUT THE INSTITUTION:

Kongu Engineering College (KEC), a distinguished Autonomous institution, is located in Perundurai, Erode district, Tamil Nadu. Established in 1984, KEC is committed to delivering value-based Education, Research, and Consultancy in Engineering and Management. The vast campus provides state-of-the-art infrastructural facilities to sculpt bright minds into ethically strong and technically competent professionals. Affiliated with Anna University and accredited with an 'A++' Grade by NAAC, the college attracts students from across India. KEC fosters a stimulating environment that promotes rigorous scholarship and community spirit. For further information about our institution, please visit URL: <https://www.kongu.ac.in/>.

ABOUT THE DEPARTMENT:

The department of Computer Science and Engineering (CSE) has been offering sound programs at both the UG and PG levels since its inception in 1988. With a cohesive team of expert faculty members, the department is dedicated to developing innovative, competent, and ethically strong computer engineers prepared for global challenges. The curriculum is regularly updated, balancing conventional and radical topics to meet the growing demands of the software industry. Core courses include Programming Languages, Computer Architecture, Networking Technologies, and new specializations like Data Science and Cyber Security.

ABOUT THE SEMINAR:

The National Level Seminar on “Next-Generation Remote Sensing and Quantum-AI Synergies for Sustainable Environmental Monitoring” focuses on emerging technologies transforming Earth observation and environmental monitoring. It highlights the integration of Artificial Intelligence with remote sensing for accurate land-use mapping, environmental change detection, and disaster monitoring. The seminar also introduces deep learning and Explainable AI to enhance prediction accuracy, transparency, and trust in data-driven systems. Additionally, it explores the potential of quantum computing for addressing complex geospatial data challenges and secure data processing. By bringing together academicians, researchers, and industry experts, the seminar promotes interdisciplinary collaboration and sustainable environmental monitoring solutions.

OBJECTIVES:

- To explore advanced AI-driven methodologies for analyzing large-scale remote sensing data to enhance environmental monitoring and decision-making.
- To understand the role of deep learning architectures in high-resolution land cover mapping, change detection, and environmental pattern analysis.
- To examine the integration of Quantum Computing with AI for solving complex geospatial and environmental challenges.
- To highlight the importance of Explainable AI (XAI) in ensuring transparency, interpretability, accountability, and trust in AI-driven environmental applications.
- To discuss AI-driven disaster monitoring and prediction frameworks.
- To promote the integration of IoT-enabled ground sensors with satellite-based remote sensing, enabling real-time, multi-source environmental data fusion.
- To foster capacity building and skill development by introducing participants to emerging tools, platforms, and research directions in AI-driven and quantum-enabled environmental monitoring for sustainable global environmental solutions.

OUTCOMES:

This seminar will enhance participants’ understanding of advanced remote sensing technologies and the role of Artificial Intelligence in environmental monitoring. It will provide insights into deep learning-based techniques for land-use mapping, environmental change detection, and disaster monitoring. The program will also introduce emerging quantum computing concepts for large-scale Earth observation systems. It will strengthen awareness of Explainable and Responsible AI, emphasizing transparency, ethical deployment, and data security. Additionally, participants will gain knowledge of AI-driven disaster prediction, real-time monitoring systems, and secure data transmission. Overall, the seminar will promote interdisciplinary learning, research-oriented thinking, and the adoption of innovative and sustainable technological solutions for effective environmental monitoring and management.



पृथ्वी विज्ञान मंत्रालय
Ministry of Earth Sciences

**MoES SPONSORED
A NATIONAL LEVEL SEMINAR
on
NEXT-GENERATION REMOTE
SENSING AND QUANTUM-AI
SYNERGIES FOR SUSTAINABLE
ENVIRONMENTAL MONITORING**

08.01.2026 & 09.01.2026

(OFFLINE)

Organized by
**DEPARTMENT OF COMPUTER SCIENCE
AND ENGINEERING**
Kongu Engineering College



सत्यमेव जयते

Ministry Of Earth Sciences
Government of India



ABOUT THE MoES:

The Ministry of Earth Sciences (MoES), Government of India, is the nodal ministry responsible for the development and implementation of programs related to weather, climate, ocean, coastal systems, seismology, and atmospheric sciences. It plays a vital role in advancing scientific research and technological innovation to improve the understanding of Earth systems and address challenges related to climate change, natural disasters, and environmental sustainability. The ministry continuously works toward strengthening scientific capabilities to support national priorities and global environmental commitments.

Through its various institutions and initiatives, MoES supports research, capacity building, and infrastructure development in areas such as weather forecasting, ocean observation, climate services, and disaster risk reduction. The ministry also focuses on strengthening observational networks, enhancing data accuracy, and promoting the use of advanced modeling and analytical tools for better environmental assessment and prediction. These efforts help improve preparedness, risk mitigation, and informed decision-making across multiple sectors.

In addition, MoES promotes interdisciplinary research, data-driven decision-making, and the application of advanced technologies to enhance societal resilience and sustainable development. It works closely with national and international organizations to improve early warning systems, support evidence-based policymaking, and build long-term capacity for managing environmental risks. Through these initiatives, the ministry plays a crucial role in safeguarding natural resources, protecting lives and livelihoods, and advancing India's environmental and scientific leadership. Furthermore, MoES actively encourages innovation, knowledge sharing, and skill development by supporting research institutions, academic collaborations, and technology-driven initiatives. By integrating science, policy, and societal needs, the ministry contributes to building a resilient and climate-ready nation, ensuring sustainable growth, environmental stewardship, and long-term socio-economic stability for future generations.

TOPICS TO BE COVERED:

- Inaugural function & Overview of Earth Observation and Remote Sensing Technologies.
- AI in Remote Sensing, Quantum Computing Fundamentals and Their Relevance to EO.
- Explainable AI (XAI) for Environmental Monitoring.
- Deep Learning for Satellite Image Classification and Change Detection.
- Sensor Fusion: Integrating Satellite, Drone, and IoT Data.
- Cloud Computing and Big Data Platforms for EO Analysis.
- Hybrid AI-Quantum Models for Geospatial Intelligence.
- Data Privacy and Cybersecurity in Remote Sensing Using Quantum Techniques & Valedictory Function.

RESOURCE PERSONS:

- Dr. Senthilkumar, Former Director, IIRS.
- Dr. S. Manthira Moorthi, Scientist-G Group Director, ODPG/SIPG, Space Applications Centre (ISRO), Ahmedabad.
- Dr. S. V. Kogilavani, Professor, National Institute of Technical Teachers Training and Research, Chennai.
- Dr. S. Malliga, Senior Professor, Department of CSE, Kongu Engineering College, Perundurai.
- Dr. M. Sethumadhavan, Director, TIFAC CORE in Cyber Security, Amrita Vidyapeetham, Coimbatore.
- Vignesh Prabhu, Senior Engineer/GenAI Architect, Bosch Global Software Technologies.
- M. Sangeetha, Professor and Head, ECE, SRM University, Chennai.

REGISTRATION

The participants are advised to apply online through the form given below.

<https://forms.gle/D3JhuoA9P3MbpggZA>

FREE REGISTRATION

for all participants

Last date of registration: 6.1.26



पृथ्वी विज्ञान मंत्रालय
Ministry of Earth Sciences

APPLY THROUGH:

<https://forms.gle/D3JhuoA9P3MbpggZA>

FREE REGISTRATION

CO-ORDINATORS

Dr. P. S. Nandhini

Associate Professor/ CSE

Dr. S. Malliga

Senior Professor and Head/CSE

P. Jeevitha

Assistant Professor/ CSE

**Dept. of Computer Science &
Engineering**

ADDRESS FOR CORRESPONDENCE:

Dr. P. S. Nandhini

Associate Professor/CSE

Kongu Engineering College,

Perundurai-638060,

Tamil Nadu, India.

Mobile: +91 99620 73299

+91 98422 78780

e-Mail: nandhini.cse@kongu.edu